

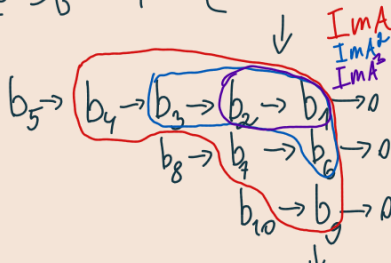
$$B = (b_1, \dots, b_{10})$$

$$[f]_B^B = A = \begin{pmatrix} J_5(0) & 0 & 0 \\ 0 & J_3(0) & 0 \\ 0 & 0 & J_2(0) \end{pmatrix} = \begin{pmatrix} \begin{matrix} b_1 & & & & \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{matrix} & \begin{matrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{matrix} & \begin{matrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{matrix} \end{pmatrix}$$

→ b_4 na b_6
 → b_8 na b_7
 → zobrazuje b_{10} na b_9
 ↓
 každý vektor b_i se zobrazuje
 buď na b_{i-1} , nebo na nulu.

$$\text{Ker}[f]_B^B = \text{Span}\{b_1, b_6, b_9\}$$

$$\text{Im}A = \text{Span}\{b_1\}$$



$$\text{Ker}A^5 \supseteq \text{Ker}A^4 \supseteq \text{Ker}A^3 \supseteq \text{Ker}A^2 \supseteq \text{Ker}A$$

$$\begin{pmatrix} b_2 \rightarrow b_1 \\ b_7 \rightarrow b_6 \\ b_{10} \rightarrow b_9 \\ \text{Ker}A^2 \supseteq \text{Ker}A \end{pmatrix} \Leftrightarrow \begin{aligned} \text{Ker}A &= \text{span}\{b_1, b_6, b_9\} \\ \text{Ker}A^2 &= \text{span}\{b_1, b_7, b_{10}\} + \text{Ker}A = \text{span}\{b_1, b_2, b_6, b_7, b_9, b_{10}\} \end{aligned}$$

$b_5 \rightarrow b_4 \rightarrow b_3 \rightarrow b_2 \rightarrow b_1 \rightarrow 0$	$\dots$	$\{b_1, \dots, b_{\min(i,5)}\}$	$\dots$	$\{b_1, \dots, b_{\max(0,5-j)}\}$	$\dots$	$k=5$
$b_8 \rightarrow b_7 \rightarrow b_6 \rightarrow 0$	$\dots$	$\{b_6, \dots, b_{5+\min(i,3)}\}$	$\dots$	$\{b_6, \dots, b_{5+\max(0,3-j)}\}$	$\dots$	$k=3$
$b_{10} \rightarrow b_9 \rightarrow 0$	$\dots$	$\{b_9, \dots, b_{8+\min(i,2)}\}$	$\dots$	$\{b_9, \dots, b_{8+\max(0,2-j)}\}$	$\dots$	$k=2$
		$i \geq k$		$j \geq k$		

kde k je počet el.-tu v řetězci

$$\dim((\text{Ker}f^i) \cap (\text{Im}f^i)) = \max(\underbrace{0, \min(i, 5-j)}_{d_1}) + \max(\underbrace{0, \min(i, 3-j)}_{d_2}) + \max(\underbrace{0, \min(i, 2-j)}_{d_3})$$

$$C = \left(\begin{cases} \emptyset \\ (b_1, \dots, b_{d_1}) \end{cases}, d_1=0 \right) \cup \left(\begin{cases} \emptyset \\ (b_6, \dots, b_{5+d_2}) \end{cases}, d_2=0 \right) \cup \left(\begin{cases} \emptyset \\ (b_9, \dots, b_{8+d_3}) \end{cases}, d_3=0 \right)$$